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# SDC Hackathon

## Phase 1 : Preliminary Project Proposal

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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**SDC HACKATHON 2026**

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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | Made in UTM |
| **Selected Domain** | Domain 1: The Aiversecity Future-Ready Workforce |
| **Selected SDG Track (1–17)** | **SDG 4** Quality Education (targets 4.3, 4.4), the domain's main SDG, with **SDG 8** Decent Work and Economic Growth (targets 8.2, 8.5) |
| **Project / Solution Name** | **Laluan** (Malay for "path"): a career GPS for the AI era |
| **Name (Leader)** | Tan Hong Sheng |
| **Email & Phone (Leader)** | tanhongsheng050204@gmail.com (phone as on the registration form) |
| **University Affiliation(s)** | Universiti Teknologi Malaysia (UTM) |
| **Number of Team Members** | 4 |

## 2. Executive Summary & Problem Statement

### 2.1 The Problem Context

*Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.*

***[Maximum 250 words]***

AI is changing which jobs exist faster than training programmes and skill frameworks can keep up. In Malaysia, 56% of professionals already use AI tools at work, yet only 26% have had formal training (Hays Asia Salary Guide 2025, cited in the Domain 1 brief), and MOSTI expects more than 600,000 workers to need reskilling within three to five years. Globally, the World Economic Forum expects a structural churn of 22% of formal jobs by 2030 (170 million created, 92 million displaced), with clerical roles shrinking fastest: data entry clerks by about 26% and administrative assistants by about 20% (Future of Jobs Report 2025, Figures 2.1 and 2.3).

Those affected first have the least room to experiment: clerks, customer-service staff and other routine-task workers, graduates choosing a first job, and HR teams deciding whom to reskill. Today they rely on static career lists that date quickly, one-off assessments, or generic chatbots whose advice cannot be traced to any source.

This matters socially and economically. A worker who reskills too late loses income and confidence; a graduate who trains for a shrinking role wastes time and money; a company that guesses wrong spends its training budget on the wrong skills. What is missing is guidance that is **personal, current, explainable and honest about uncertainty**: it should show which skills carry over, which roles are likely to grow under different AI-adoption futures, and an affordable plan to get there, without making the decision for the person.

### 2.2 The Proposed AI Solution

*Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.*

***[Maximum 300 words]***

Aisyah, a data-entry clerk, learns her team adopted an AI invoice reader. Laluan takes her from “who I am today” to “where I can be in the AI era” in minutes.

1. **Skills in.** CV upload, questionnaire or skill list. Names, IC and phone numbers are removed *before* any AI reads the text; the file is never stored. She confirms the skills.

2. **Forecasts, not lists.** A re-runnable pipeline collects three years of job-ad mentions, AI-research papers (arXiv) and Malaysian employment (OpenDOSM). Each series becomes a damped trend to 2030 with an 80% range, flagged *emerging* when growth speeds up. A backtest on the last year is shown in the app. Every refresh is dated and **only goes live after a person approves it**.

3. **Three futures.** Slow, baseline and accelerated AI adoption change the forecasts and her ranked roles: 0.6 × skill fit + 0.4 × outlook. "Why this rank?", a step-by-step forecast x-ray and **tipping points** (the adoption speed at which her job turns declining) explain every number.

4. **A plan she can start this week.** Ordered courses for missing skills, with hours, RM price and HRD Corp status from a curated catalogue; AI summaries are checked against it, so nothing is invented.

5. **SDG links with reasons** for every role and skill, and an **SDG lens** showing what choosing for a goal costs in hours and fit; an **AI coach** that only suggests, **automatic re-planning**, a public **fairness check** and an **HR view**.

Hardest risk: reliable, updatable trend data. Fallbacks: noisy trends shrink towards no change, ranges stay visible, arXiv falls back to OpenAlex, and the last approved snapshot stays live.

Workshops used: **GenAI APIs** (Gemini; Qwen2.5 and Whisper on RunPod; offline fallback), **database architecture** (Supabase Postgres) and **cloud deployment** (Vercel, Docker on Render, GitHub Actions).

## 3. Technical Architecture & Implementation Plan

### 3.1 Conceptual Tech Stack

*List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.*

*Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.*

**Frontend:** React 18 + TypeScript + Vite; responsive; EN / BM / 中文; lite mode for slow connections

**Backend API:** Python 3.12, FastAPI, Pydantic, in a Docker container

**AI:** Google Gemini API (gemini-3.8-flash for text, gemini-embedding-001 for embeddings); open models on RunPod Serverless: Qwen2.5-7B-Instruct (CV reading, so raw CV text stays away from Google), faster-whisper small (voice input) and our own OCR worker (scanned CVs); rules and templates when no key is set

**Data:** Supabase Postgres and Auth (profiles, progress, Future Cards, trend snapshots, review decisions; optional Google or Microsoft login); versioned JSON (roles, skills, courses, SDG map, O*NET tasks)

**Trend pipeline:** arXiv with OpenAlex fallback (AI research); Malaysian job ads (freehire.me); Hacker News "Who is hiring?" threads; developer attention (Stack Overflow, GitHub, npm, PyPI); OpenDOSM labour force, sector employment, prices and salaries. Three years of history per series → damped-trend forecasts to 2030 with 80% ranges and a backtest (plain Python)

**Hosting:** Vercel (frontend), Render (API, Singapore), Supabase (Singapore), RunPod Serverless (open models), GitHub Actions (tests on every push, weekly trend refresh)

System block diagram: see the Appendix.

**Sponsor resources: Yes. RunPod** (Domain 1 sponsor): Serverless endpoints run Qwen2.5-7B-Instruct for CV reading, faster-whisper for voice input and our own OCR worker for scanned CVs, so raw CV text and audio never reach Google. **Replit** (Domain 1 sponsor): development workspace and an alternative host for the same code. Gemini is used on Google's free tier; Vercel, Render and Supabase are on free plans.

### 3.2 Design-Thinking & Deployment Strategy

*Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.*

***[Maximum 200 words]***

**Empathise.** Personas from the brief's users: Aisyah, a clerk facing an AI invoice reader; Wei Jie, a final-year student; an HR team planning reskilling. Their fears set the tone: calm, honest, never alarmist.

**Define.** A ranked, explainable plan in under five minutes, every number traceable to a dated source.

**Ideate.** We rejected a pure chatbot because its answers cannot be checked. AI reads CVs and explains; forecasts come from public data with a published method.

**Prototype and test.** Live since Phase 1. Browser testing fed back into the design: a phone layout, a "server waking up" notice, human approval of trend data. A backtest exposed that plain trend lines overstate change, so we added shrinkage and damping. Every fix is in REVIEW_LOG.md.

**Deploy.** Vercel (frontend), Docker on Render (API), Supabase (data), tests on every push. It works without an API key. Phase 2: human evaluation with a counsellor and native speakers via the in-app review kit.

## 4. Preliminary Environmental & SDG Impact Framework

*To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.*

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

| Metric Category | Your Response |
|---|---|
| **Target SDG Sub-Target(s)**<br>*Identify the exact UN SDG sub-target your project addresses*<br>*e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."* | **4.4** "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship."; **8.5** "By 2030, achieve full and productive employment and decent work for all women and men, including for young people…". Supporting: 4.3 (affordable technical and vocational education), 8.2 (productivity through innovation). |
| **Current Problem Baseline**<br>*What is the starting condition of the problem, expressed as a number?*<br>*e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."* | Only **26%** of Malaysian professionals using AI at work have had formal AI training (Hays Asia Salary Guide 2025, as cited in the Domain 1 brief); **600,000+** Malaysian workers need reskilling within 3–5 years (MOSTI, same source); data-entry clerk jobs are projected to fall **≈26%** by 2030 (WEF Future of Jobs Report 2025, Fig. 2.3). For an individual today: no source-cited, personalised pathway exists in one place, so the share of recommendations a user can trace to a source is effectively **0%** in generic chatbot advice. |
| **Core Measurable KPI**<br>*The specific unit you will track*<br>*e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.* | (1) Time from CV to a ranked, costed pathway (minutes). (2) Skill-extraction recall on 5 synthetic CVs marked by a person (%). (3) Share of role, skill, task and course recommendations that show a dated source (%). (4) Share of plan steps that are free or free to audit (%). (5) Reviewer agreement with the top-3 roles for the two personas (%). (6) Forecast backtest: share of clear skill trends whose direction held in the held-out year (%). |
| **Phase 2 Impact Target**<br>*Your realistic, percentage-based target by the final prototype presentation*<br>*e.g., "20% improvement in computational* | (1) ≤ **5 minutes** in a timed run. (2) ≥ **80%** average recall. (3) **100%** of recommendations cited (already enforced by automated tests). (4) ≥ **70%** of steps free or free to audit in both demo pathways (currently 2 of 2 for Aisyah and 1 of 1 for Wei Jie). (5) ≥ **70%** reviewer agreement from a lecturer or career counsellor. (6) Above the "everything rises" guess, re-measured at every refresh (21 Oct 2026: 14 of 19 clear trends |

| Metric Category | Your Response |
|---|---|
| *resource use."* | held, against 12 for "everything rises"; sizes still rougher than a "no change" guess).<br><br>**Sources:** World Economic Forum, *Future of Jobs Report 2025*, January 2025, Figures 2.1 and 2.3, https://reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf. SDC Hackathon, *Domain 1 Problem Statement & Marking Rubric* (citing Hays Asia Salary Guide 2025 and MOSTI). United Nations, *SDG targets 4.3, 4.4, 8.2, 8.5*, https://sdgs.un.org/goals. |

## 5. Eligibility & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

## 6. IP & Originality Declaration

### 6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it: Not applicable. All code, design and research were created during the SDC 2026 hackathon period.

### 6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

| Name | Type (code/dataset/model) | License |
|---|---|---|
| React 18.3.1, React DOM 18.3.1 | Code | MIT |
| Vite 5.4.11, @vitejs/plugin-react 4.3.4 | Code | MIT |
| TypeScript 5.6.3 | Code | Apache-2.0 |
| qrcode 1.5.4 | Code | MIT |
| @supabase/supabase-js 2.45.4 | Code | MIT |
| Inter font (via @fontsource-variable/inter 5.3.0) | Font | SIL OFL 1.1 |
| FastAPI, Starlette, Pydantic | Code | MIT / BSD-3 / MIT |
| Uvicorn, HTTPX, pypdf | Code | BSD-3 |
| psycopg 3, PyJWT | Code | LGPL-3.0 / MIT |
| RapidOCR, ONNX Runtime, pypdfium2, Pillow (our own OCR for | Code | Apache-2.0 / MIT / Apache-2.0 or BSD-3 / MIT-CMU |

| | | |
|---|---|---|
| scanned CVs) | | |
| python-multipart, google-genai SDK, runpod SDK | Code | Apache-2.0 / Apache-2.0 / MIT |
| PyMuPDF (only for the one-off WEF figure extraction script, not in the app) | Code | AGPL-3.0 |
| PaddleOCR PP-OCRv4 English text recogniser and RapidOCR's text detector | Model | Apache-2.0 |
| Google Gemini gemini-3.8-flash, gemini-embedding-001 (called via API) | Model | Gemini API terms |
| Qwen2.5-7B-Instruct (Alibaba Qwen team), served with vLLM on RunPod Serverless | Model | Apache-2.0 |
| OpenAI Whisper small via faster-whisper, on RunPod Serverless | Model | MIT |
| O*NET 30.0 task statements (U.S. Department of Labor) | Dataset | CC BY 4.0 |
| WEF Future of Jobs Report 2025, Figure 2.3 (figures cited, not redistributed) | Dataset | WEF terms of use, with citation |
| UN Sustainable Development Goals names, targets and colours | Dataset | UN SDG guidelines |
| arXiv API metadata counts; OpenAlex | Dataset | arXiv API terms; CC0 |
| OpenDOSM labour force, employment, prices and salaries (via data.gov.my) | Dataset | CC BY 4.0 |
| Hacker News “Who is hiring?” posts, counted via the Algolia HN API | Dataset | HN API terms (counts only) |
| Malaysian job postings via the freehire.me API | Dataset | freehire terms; project MIT |
| Stack Overflow, GitHub, npm and PyPI activity counts | Dataset | Each API's terms (counts only) |
| Anthropic Economic Index | Dataset | CC BY |
| OpenAI “GPTs are GPTs” occupation exposure scores | Dataset | MIT |
| ILO Working Paper 140, Annex Table A1 | Dataset | CC BY 4.0 |
| Three landing-page photos from Unsplash (Gema Saputera, Mimi | Images | Unsplash License |

### 6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns: Not applicable.

## 7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature: Tan Hong Sheng (Team Leader)

Date: 24 October 2026

# Appendix: System Block Diagram

